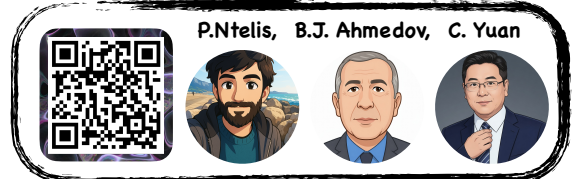


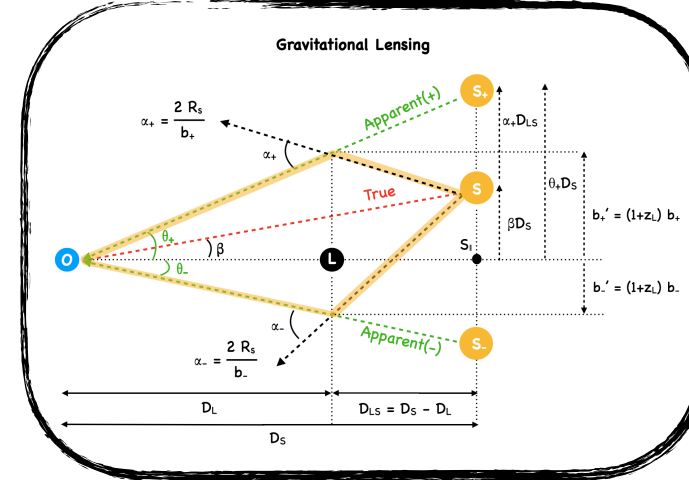
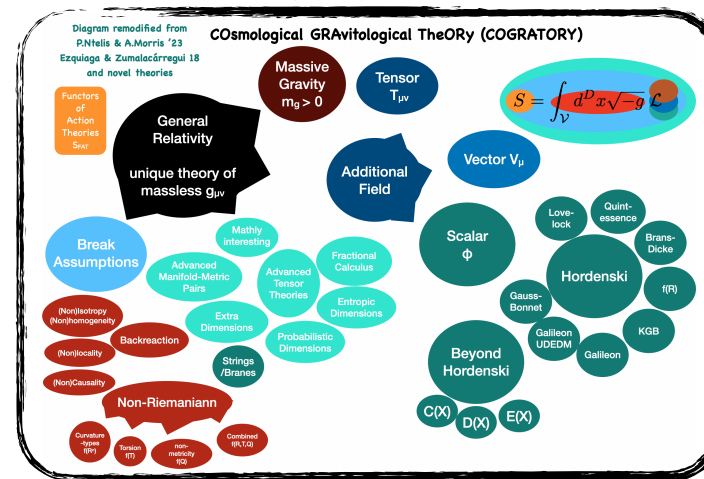
# New math structures in cosmology:

## A cosmic femtolensing Hawking radiation sign



We propose a novel formalism to detect signatures of Hawking radiation (HR) from primordial black holes (PBHs) in the mass range of  $10^{-16}$  to  $10^{-13} M_{\odot}$  using the femtolensing effect, within the framework of dark messengers. By adapting the micro- and pico- lensing frameworks for gamma-ray bursts, we model the femto-lensing of HR, characterized by a thermal spectrum in the eV-GeV range, by PBHs acting as gravitational lenses. We derive the magnification and temporal signatures of femtolensed HR, accounting for the small Einstein radii and short timescales inherent to femtolensing. The interplay between the lensed flux and spectral distortions offers a unique probe of PBH evaporation. We discuss observational strategies using current and future telescopes sensitivities which potentially can detect these signatures, and provide constraints on PBH populations and their contribution to dark matter. Furthermore, we connect the idea of femtolensing Hawking radiation to dark energy, through estimates of the cosmological distances, and energy densities of the universe. This can lead to a novel observational signature of dark energy, the evolution of the universe, through HR, and PBH, and potentially change our understanding about cosmology.

Thank you, 谢谢, Xiè xiè, Ευχαριστώ ! :)



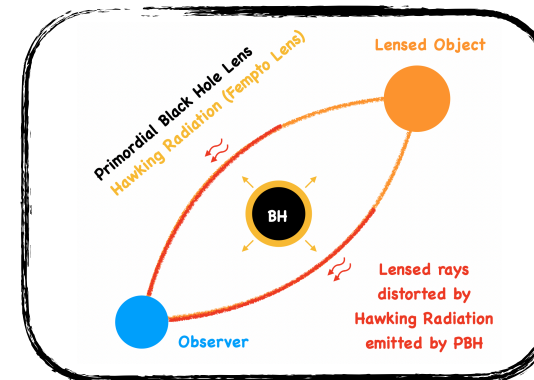
Universions



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The flux of photons per second, per meter square is proportional to lens redshift, to mass ratio of lens with the source, to the inverse of the emitter mass, inverse square of the 1 plus redshift, and inverse square to the comoving distance of the observer from the emitter:

$$\frac{\Phi_{\text{HR, lensed}}(z_S, z_L, M_L, M_S, H_0, \Omega_{m0})}{\text{ph s}^{-1} \text{m}^{-2}} \approx \frac{1.4 (1+z_L) \left(\frac{M_L}{M_S}\right)}{1 - e^{-1.4 (1+z_L) \left(\frac{M_L}{M_S}\right)}} \times 6 \times 10^{30} \left(\frac{M_S}{\text{kg}}\right)^{-1} \left[\frac{(1+z_S) d_C(z_S, H_0, \Omega_{m0})}{\text{m}}\right]^{-2}$$



Conference  
Dark Matter & Stars:  
Multi-Messenger  
Probes of Dark Matter  
and Modified Gravity  
13-15 Jul 2026 @  
Southampton  
University



- A cosmic femtolensing Hawking radiation

-  $\Lambda$ CDM (in cosmic observations) (Talk)



- Functors of Actions Theories (FAT) (Cosmos APP)

- Quintessence



- Extra Dimensions in Observations

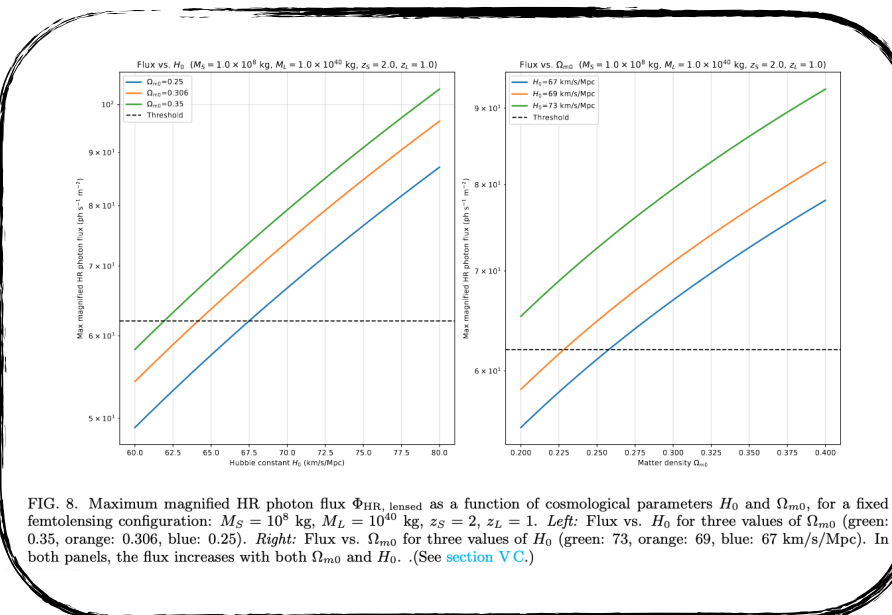
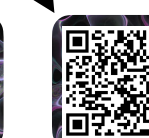
- Advanced Manifold-Metrics Pairs (AMMP)



- A temporal-spatial-entropic K curvature

- EinsteinKPy2Tex code

- Advanced Tensor Theories (ATT)



We identify the following 4 cases of the phenomenon of the lensing effect, combined with Hawking radiation:

- The light travels from the lensed object towards around the black hole lens
  - without HR and then it travels towards the observer.
  - with HR and then it travels towards the observer.
- The light travels from the lensed object with HR towards around the black hole lens
  - without HR and then it travels towards the observer.
  - with HR and then it travels towards the observer.